

SOLUTION REQUIREMENTS

Date	15 March 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

This document defines the functional and non-functional requirements for the Voice-Based Concept Understanding Analyzer. Functional requirements describe what the system should do, while non-functional requirements specify quality, performance, security and usability expectations.

Step 1 : Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Secure user login using username/password and session management.	High
2	Authorization Levels	Role-based access for Admin and User.	High

3	External Interfaces	Integrates with Whisper/OpenAI or Gemini APIs and PDF libraries.	High
4	Transaction Processing	Upload audio, process transcription, analyze speech and generate reports.	High
5	Reporting	Generate downloadable PDF reports with similarity score and feedback.	High
6	Business Rules	Only supported audio formats are accepted. Scores are calculated automatically.	Medium
7	Compliance	Protect uploaded data and maintain user privacy.	High
8	Other	Provide dashboard, history and report storage.	Medium

Step 2 : Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast transcription and report generation.	Response within 5 seconds (excluding model latency).
2	Scalability	Support multiple concurrent users.	100+ concurrent users.
3	Security & Data Privacy	Protect uploaded audio and reports.	Encrypted storage and secure communication.
4	Reliability & Availability	System should remain available.	99% uptime.
5	Usability & Accessibility	Responsive interface with easy navigation.	Works on desktop, tablet and mobile.

6	Other	Maintainability and portability.	Modular Python code and reusable components.
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